JPE Replication Report

2026-09-09

1 Paper Identification

ID	20240098
Author	Sollaci
Title	Managers and Productivity in Retail
Iteration	4

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

3 General

- ☒ Data Availability and Provenance Statements
  - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - ☐ License for Data (optional, but recommended)
  - ☐ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements

    - ☒ Software Requirements
    - ☒ Controlled Randomness (as necessary)
    - ☒ Memory
    - ☒ Runtime
    - ☐ and Storage Requirements

- ☒ Description of programs/code
    - ☐ License for Code (Optional, but recommended)
- ☐ {REPLICATOR} to Replicators
    - ☐ Details (as necessary)
- ☐ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
    - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
    - ☐ A PDF copy of IRB approval is contained in the package.
    - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
    - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
    - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
    - ☐ A PDF copy of IRB approval is contained in the package.
    - ☐ A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- ☒ is a simulation study: the data are generated by an algorithm. (Parts of it.)
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

The data was provided to the authors under the condition that the companies the data stems from remain anonymous. The data was provided for replication purposes but data sources, references and access conditions were not provided for this reason.

The authors specified that the data "should be preserved in the restricted-access location of the Journal of Political Economy for as long as necessary for the successful replication of the paper. It should be deleted after the replication is confirmed. The authors commit to preserving the data and code for a period of at least five years following the publication of the paper and to provide reasonable assistance to requests for clarification and replication."

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README:

- The authors provided a codebook for the data describing the variables contained in the datasets.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
data/A_dataset.dta	no	yes	no	no	no	no
data/B_dataset.dta	no	yes	no	no	no	no

5.1.1 Data

Data are not cited, no access description provided either, no information regarding the data is provided.

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided, and works: http://data.bls.gov/cgi-bin/surveymost?sm+08
- ☐ Access conditions are described:
  - No information on access conditions.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☐ Data are cited
  - ☐ in the manuscript
  - ☐ in the README:

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☒ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper

  {REQUIRED} Please mention the authors affiliations

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
  - R (version 4.4.0)
  - Stata (version 19)
  - Python (version 3.11.11)
- ☒ Computational Requirements specified as follows:
  - R: packages: haven, rstudioapi, doBy, lfe, Matrix, RhpcBLASctl
  - Stata: packages to be installed: reghdfe, ftools, did_imputation, event_plot, binscatter, grc1leg2, schemepack
  - Python: libraries:numpy, pandas, scipy, matplotlib, os
  - ebayes.ado and fese_fast.ado (provided in repository)
- ☒ Time Requirements specified as follows:
  - ca. 2h
- ☒ Requirements are complete.

For missing requirements, see the list of required changes in Section 9

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 0
- Variables flagged in data: 0
- Code files with PII references: 74
- PII references in code: 7693

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Code	`01_main.R`	5	name, loc
Code	`02_main.do`	12	son, lat, name, gender
Code	`03_main.sh`	2	lat
Code	`A_analyze_FE.do`	10	name, lat, loc, location, lon
Code	`A_estimate_fe.R`	8	network, name
Code	`A_estimate_gfe.R`	8	network, name, lat
Code	`A_event_study.do`	80	loc, name, lon, son, lat
Code	`A_prelim_analysis.do`	10	loc, location, lat, name
Code	`A_revenue_counterfactuals.do`	8	loc
Code	`B_analyze_FE.do`	6	name, lat, loc, location
Code	`B_estimate_fe.R`	8	network, name
Code	`B_estimate_gfe.R`	9	lat, network, name
Code	`B_event_study.do`	80	loc, name, lon, son, lat
Code	`B_gender_moves.do`	7	gender, lon, loc, location
Code	`B_prelim_analysis.do`	11	name, lat
Code	`B_revenue_counterfactuals.do`	8	loc
Code	`_eststo.ado`	8	loc, lon
Code	`binscatter2.ado`	209	lat, name, loc, lon, lname, block, coord
Code	`calibrate_NAM.py`	1	lat
Code	`create_table_1_clean_from_fe.R`	11	name, block, loc, lon, lname

See Appendix for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-08-15T19:39:36.087

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
79	11	13	2	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

- Folder A contains 2 R scripts, a master Stata do-file which references to 5 seperate stata files. Additionally it contains 2 python files and another separate stata do-file for the simulation part of the study. Folder B replicates the same structure of folder A for the analysis for company B instead of company A, excluding the simulation.

cloc	github.com/AlDanial/cloc v 2.06 T=0.14 s (797.6 files/s, 378370.9 lines/s)

Language	files	blank	comment	code
Stata	99	7742	3483	40487
R	6	236	82	638
Python	2	77	49	173
TeX	3	0	0	145
Text	1	0	0	13
Bourne Shell	1	2	1	4
SUM:	112	8057	3615	41460

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

- MacBook Pro 24Go RAM M4Pro
- R 4.5.1
- Stata/MP 19
- Positron with Python 3.14

Replication time : ~33 mn (25mn for R/7mn for Stata/<1mn for Python)

and

Nuvolos 4 CPU threads and 16Go RAM - R 4.4.1 - Stata/MP 18 - VSCode

Replication time : 1h05 (51 mn for R/15 mn for Stata/<1mn for Python)

8 Replication

- Downloaded code and data from dropbox.
- Commited code to git repo.
- Installed all R packages mentioned in the ReadMe. Precise to change the `proj_root` in `01_main.R`
- Installed all Stata packages mentioned in the ReadMe.
- Imported all Python libraries as per ReadMe.
- Running R scripts on MacBookPro and Nuvolos : ok.

- Running Stata scripts on MacBookPro and Nuvolos : ok.
- Running Python scripts on MacbookPro and Nuvolos without issue.

9 Findings

It seems that table 1 is not outputed directly but must be written inline in the latex file. I generated a code outputing the table `create_table_1_clean_from_fe.R`. Same issue for table 2, I did not produce compiling code.

9.1 Missing Requirements

- ☐ Software Requirements
  - ☒ Stata
    - ∗ package: require (was missing from the required packages list)
  - ☒ R
  - ☒ Python
    - ☐ Packages Version (missing)
- ☒ Computational Requirements specified as follows:
  - Lenovo Thinkpad T14s, AMD Rizen AI 7 PRO with Radeon 860M, 32Go of RAM
- ☒ Time Requirements
  - ~1 hour for complete reproduction

  [REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

`A_estimate_fe.R` prepares the following datasets: `A_connected.dta`, `A_manager_fe.dta`, `A_store_fe.dta`.

`A_estimate_gfe.R` prepares the following datasets: `A_manager_gid.dta`, `A_manager_gfe.dta`, `A_store_gid.dta`, `A_store_gfe.dta`.

`A_fixed_effects.do` prepares `A_dataset_fe_eb.dta`.

`A_grouped_fixed_effects.do` prepares `A_dataset_gfe.dta`.

`A_event_study.do` prepares `A_event_study.dta`.

`B_estimate_fe.R` prepares the following datasets: `B_connected.dta`, `B_manager_fe.dta`, `B_store_fe.dta`.

`B_estimate_gfe.R` prepares the following datasets: `B_manager_gid.dta`, `B_manager_gfe.dta`, `B_store_gid.dta`, `B_store_gfe.dta`.

`B_fixed_effects.do` prepares `B_dataset_fe_eb.dta`.

`B_grouped_fixed_effects.do` prepares `B_dataset_gfe.dta`.

`B_event_study.do` prepares `B_event_study.dta`.

`selection_based_NAM.do` prepares `simulate_match.dta`.

9.3 Tables and Figures

- Table 2: The numbers reproduced do not match the table in the manuscript. See screenshots below:

	manager...	store_bin	freq	lab_man...	lab_store
1	1	1	111	Top	Top
2	2	1	232	Middle	Top
3	3	1	461	Bottom	Top
4	1	2	227	Top	Middle
5	2	2	333	Middle	Middle
6	3	2	289	Bottom	Middle
7	1	3	456	Top	Bottom
8	2	3	321	Middle	Bottom
9	3	3	173	Bottom	Bottom

(a) Reproduced output should correspond to the company A side of table 2

	manager...	store_bin	freq	lab_man...	lab_store
1	1	1	32	Top	Top
2	2	1	49	Middle	Top
3	3	1	88	Bottom	Top
4	1	2	51	Top	Middle
5	2	2	66	Middle	Middle
6	3	2	78	Bottom	Middle
7	1	3	97	Top	Bottom
8	2	3	72	Middle	Bottom
9	3	3	61	Bottom	Bottom

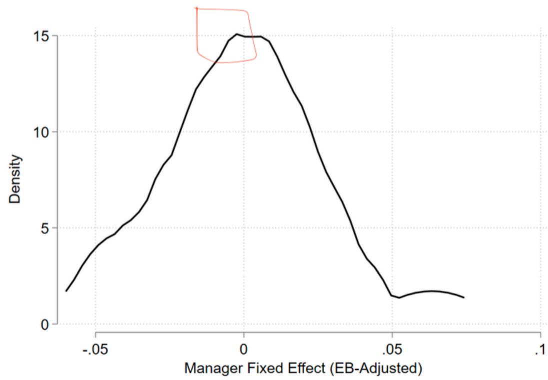
(b) Reproduced output should correspond to the company B side of table 2

Table 2: Number of manager and store matches

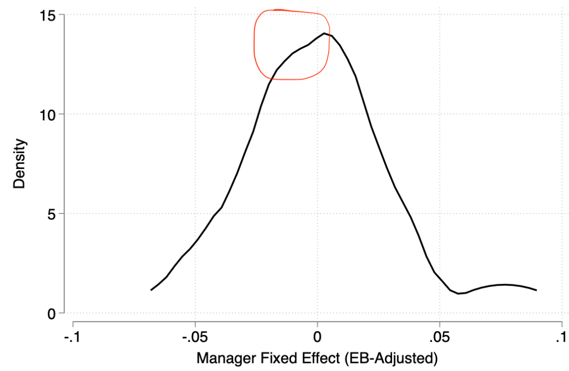
		Stores					
		Company A			Company B		
		Top	Middle	Bottom	Top	Middle	Bottom
Managers	Top	33	51	97	116	229	461
	Middle	49	67	69	225	332	285
	Bottom	88	76	64	457	322	173

(c) Paper version

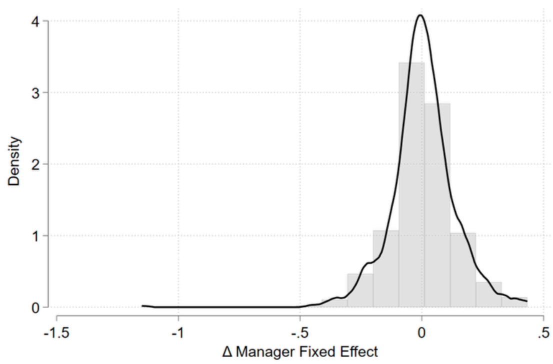
- Figure 2: looks the same.
- Figure 3: Figures slightly diverges (example for Figure 3, subfigure a Manager FE in Company A)
- Figure 4: Looks the same.
- Figure 5: looks the same
- Figure 6: looks the same.
- Figure 7: diverges.
- Table 3: Numbers diverge similar as table 2.
- Figure 8: results strongly diverge.
- Figure 9: looks the same.



(a) Paper Output

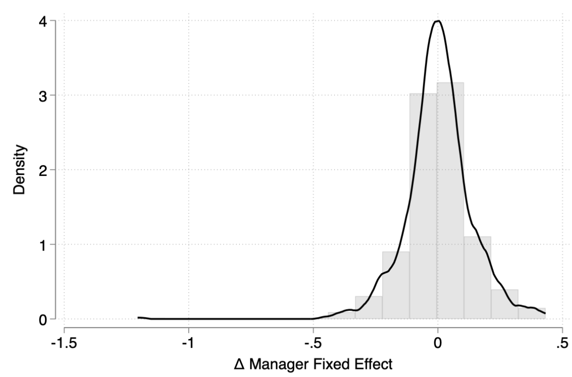


(b) Reproduced Output



(b) Manager FE (Company B)

(a) Paper Output



(b) Reproduced Output

	start_in	move_to	freq	lab_start	lab_go	total_mng	pct	pct_round
1	1	1	4	Top	Top	41	9.756098	10
2	2	1	25	Middle	Top	49	51.02041	51
3	3	1	25	Bottom	Top	74	33.78378	34
4	1	2	20	Top	Middle	41	48.78049	49
5	2	2	5	Middle	Middle	49	10.20408	10
6	3	2	33	Bottom	Middle	74	44.59459	45
7	1	3	17	Top	Bottom	41	41.46341	41
8	2	3	19	Middle	Bottom	49	38.77551	39
9	3	3	16	Bottom	Bottom	74	21.62162	22

(a) Reproduced output should correspond to the company A side of table 3

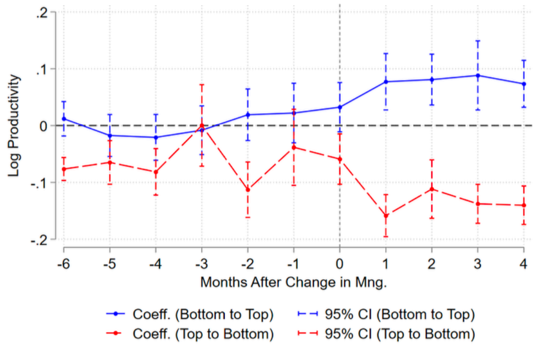
	start_in	move_to	freq	lab_start	lab_go	total_mng	pct	pct_round
1	1	1	80	Top	Top	233	34.33476	34
2	2	1	82	Middle	Top	265	30.9434	31
3	3	1	80	Bottom	Top	309	25.88997	26
4	1	2	93	Top	Middle	233	39.91416	40
5	2	2	76	Middle	Middle	265	28.67924	29
6	3	2	113	Bottom	Middle	309	36.56958	37
7	1	3	60	Top	Bottom	233	25.75107	26
8	2	3	107	Middle	Bottom	265	40.37736	40
9	3	3	116	Bottom	Bottom	309	37.54045	38

(b) Reproduced output should correspond to the company B side of table 3

Table 2: Number of manager and store matches

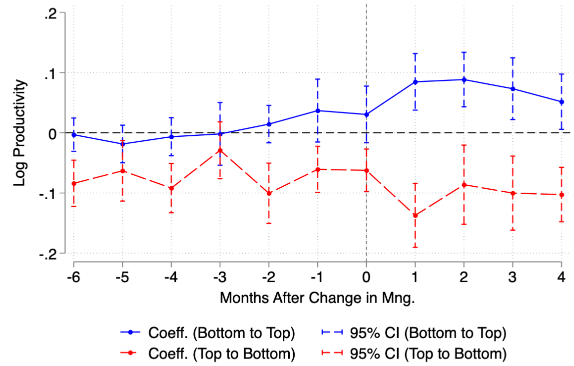
		Stores					
		Company A			Company B		
		Top	Middle	Bottom	Top	Middle	Bottom
Managers	Top	33	51	97	116	229	461
	Middle	49	67	69	225	332	285
	Bottom	88	76	64	457	322	173

(c) Paper version

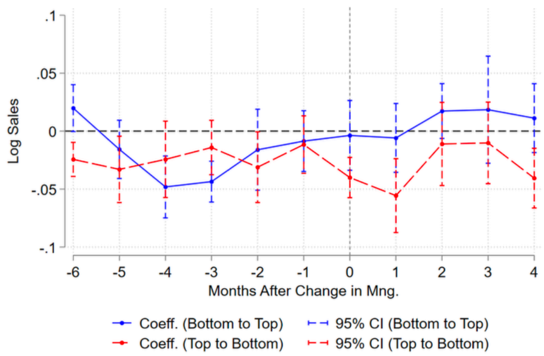


(a) Productivity (log)

(a) Paper Figure 8 A

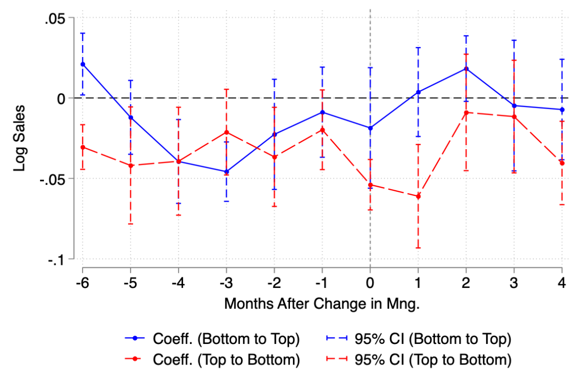


(a) Reproduced Output

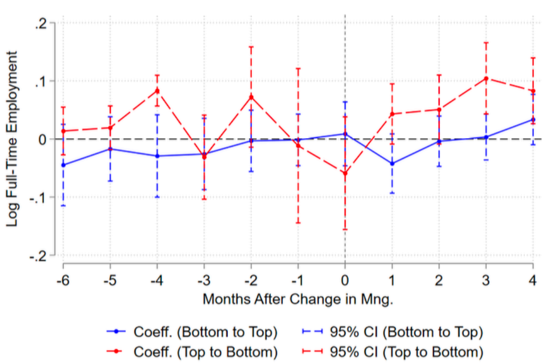


(b) Sales (log)

(a) Paper Figure 8 A

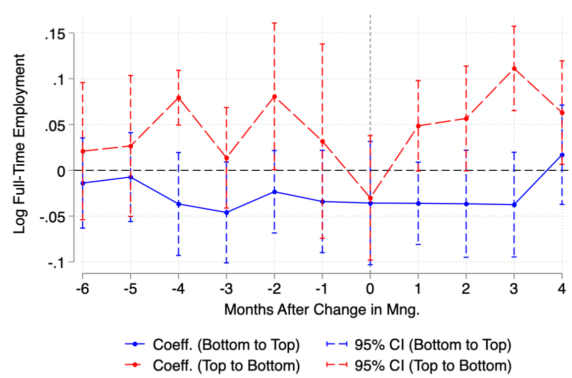


(a) Reproduced Output

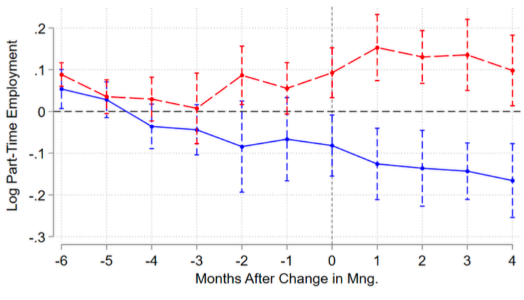


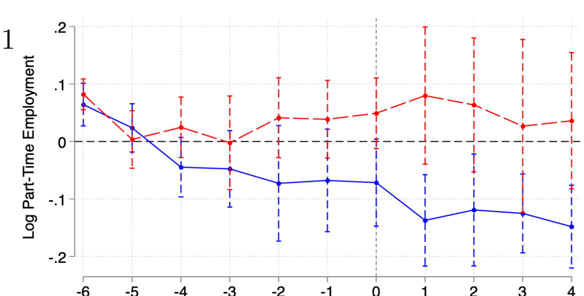
(c) Full-Time Employment (log)

(a) Paper Figure 8 A



(a) Reproduced Output





- Table 4 results for effect of switching from top to bottom manager are halved compared to the paper estimation.

Panel A: Productivity Effects β_k Before and After Managerial Switch

	Company A			
	Top to Bottom		Bottom to Top	
	Before Change	After Change	Before Change	After Change
Mean	-0.062	-0.121	0.001	0.070
Standard Deviation	0.039	0.039	0.019	0.022
Difference in Means	-0.059		0.069	

(a) Paper Table 4 for Company A

stat	tb_before	tb_after	bt_before	bt_after
Mean	-0,071	-0,098	0,004	0,066
Standard De	0,026	0,027	0,019	0,024
Difference i	-0,026		0,062	

(a) Reproduced Table 4 for Company A

	Company B			
	Top to Bottom		Bottom to Top	
	Before Change	After Change	Before Change	After Change
Mean	0.008	-0.091	0.017	0.121
Standard Deviation	0.015	0.063	0.010	0.064
Difference in Means	-0.099		0.105	

(a) Paper Table 4 for Company B

stat	tb_before	tb_after	bt_before	bt_after
Mean	0,008	-0,087	0,018	0,119
Standard De	0,014	0,064	0,009	0,062
Difference i	-0,095		0,101	

(a) Reproduced Table 4 for Company B

Panel B: Change in (EB-adjusted) Manager Fixed Effects After Managerial Switch

	Company A		Company B	
	Top to Bottom	Bottom to Top	Top to Bottom	Bottom to Top
Mean	-0.053	0.061	-0.101	0.128
Standard Deviation	0.053	0.067	0.083	0.108
Number of Changes	41	40	112	128

(a) Paper Table 4 Alternative Panel for Company A and B

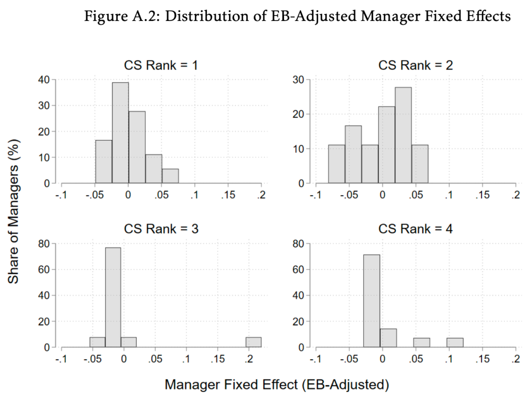
stat	top_to_bott(	bottom_to_top
Mean	-0,052	0,060
Standard De	0,051	0,067
Number of (	43,000	43,000

(a) Reproduced Table 4 Alternative Panel for Company A

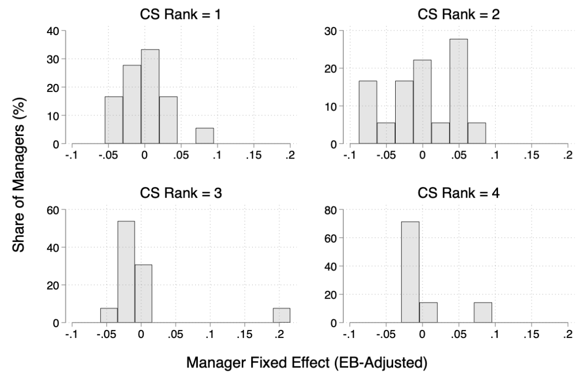
stat	top_to_bott(	bottom_to_top
Mean	-0,101	0,124
Standard De	0,083	0,105
Number of (	111,000	130,000

(a) Reproduced Table 4 Alternative Panel for Company B

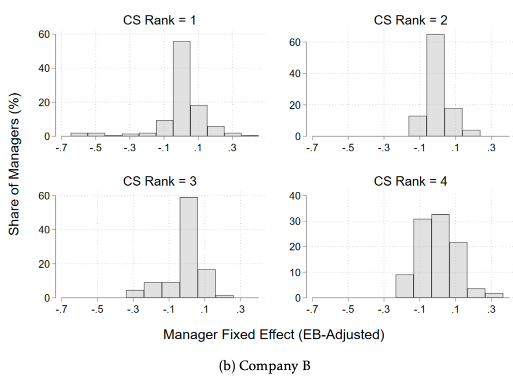
- Figure A1 looks the same.
- Figure A2 is showing discrepancies:
- Figure A3 showcases similar discrepancies as figure A2.
- Figure A4 looks the same.
- Figure A5: Looks the same.
- Table B.1 looks the same.



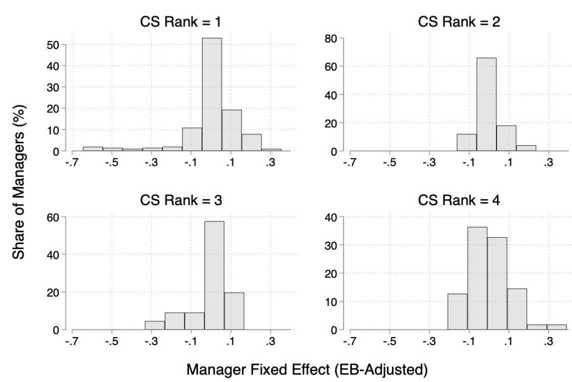
(a) Paper version Company A



(b) Reproduced figure A2 Company A

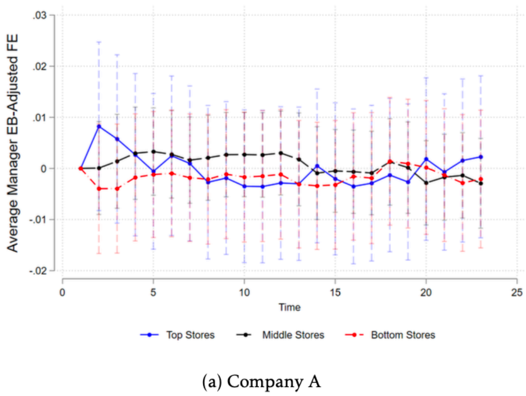


(a) Paper version Company B

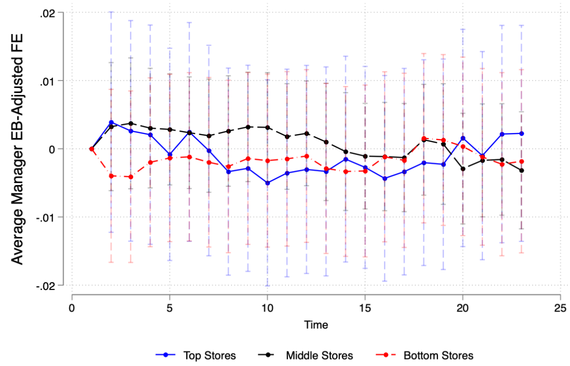


(b) Reproduced figure A2 Company B

- Table B.2 looks the same.
- Table C.1 looks the same.
- Figure C.1 looks the same.
- Figure C.2 looks the same.
- Figure C.3 looks the same.
- Table D1 are similar.
- Figure D1 displays discrepancies for both companies:

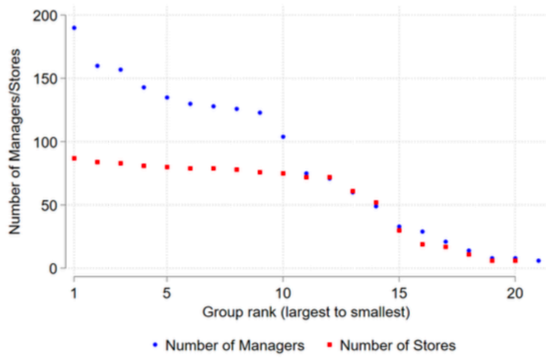


(a) Paper version Company A



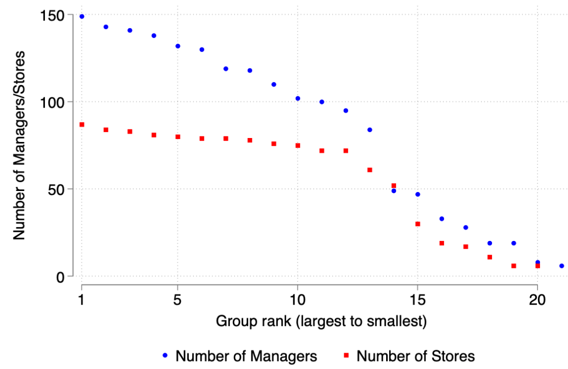
(b) Reproduced figure D1 Company A

- Figure D2 looks the same.
- Figure D3 showcases some discrepancies, but the figures are based on a randomised simulation, which could explain reasonably minor differences:
- Figure E1: diverge for company A
- Figure E2: diverges for both firms.
- Figure E3 look the same.
- Table F.1 shows discrepencies :
- Figure G.1: looks the same.
- Figure G.2: looks the same.
- Figure G.3: discrepancy in (c) full-time employment.
- Figure G.4: looks the same.



(a) Company A

(a) Paper for Company A



(a) Reproduction for Company A

- Figure G.5: looks the same.
- Figure G.6: looks the same.

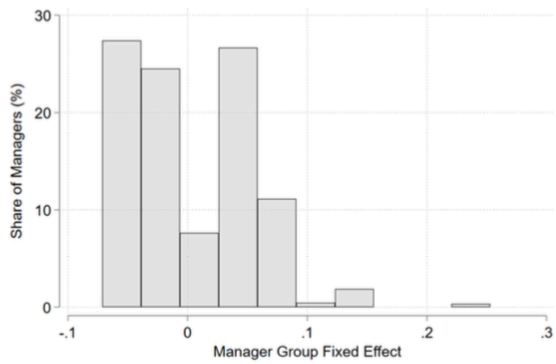
9.4 In-Text Numbers

> {REPLICATOR}: list page and line number of in-text numbers. If ambiguous, cite the surrounding text, i.e., "the rate fell to 52% of all jobs: verified".

All in-text numbers have been justified by the authors in the README.

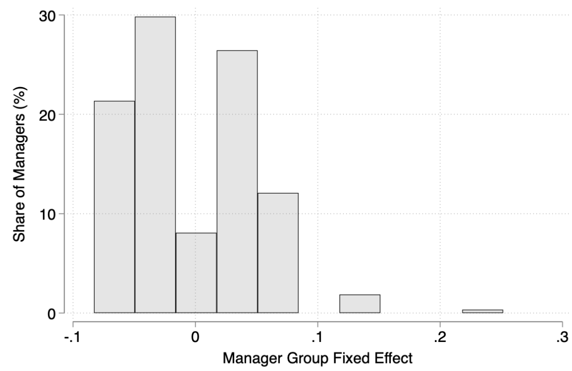
9.5 Other Issues

- Line of code where results are reproduced in the codes are not provided in the ReadMe.
- There is no clear mapping of tables/figures in the manuscripts to the tables and figure names in the results folder. Many tables and results are not saved in the results folder which makes the replication time consuming. The lines of codes were tables and figures are reproduced need to be checked in the code files.
- There is no systemic output of tables as they are shown in the paper.

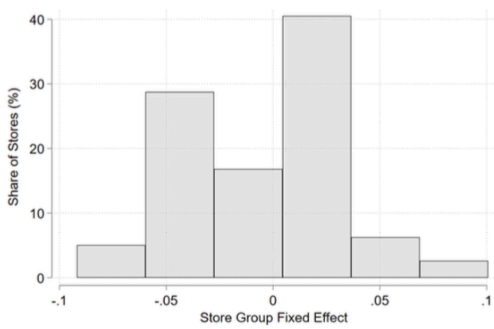


(a) Manager Fixed Effect (Company A)

(a) Paper for Company A

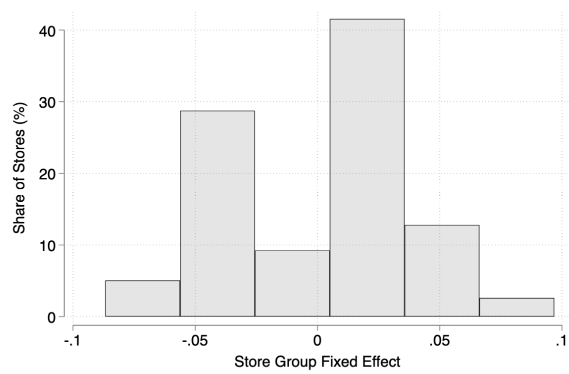


(a) Reproduction for Company A

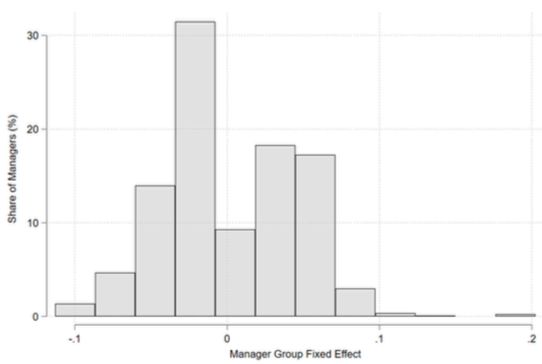


(b) Store Fixed Effect (Company A)

(a) Paper for Company A

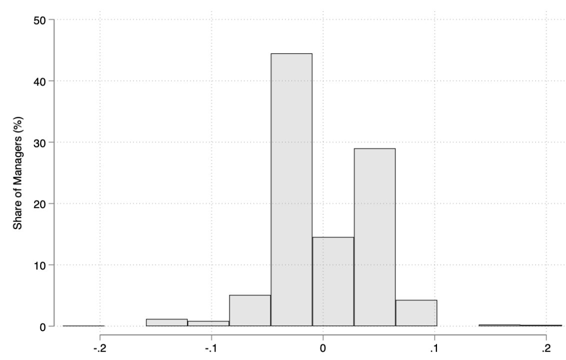


(a) Reproduction for Company A

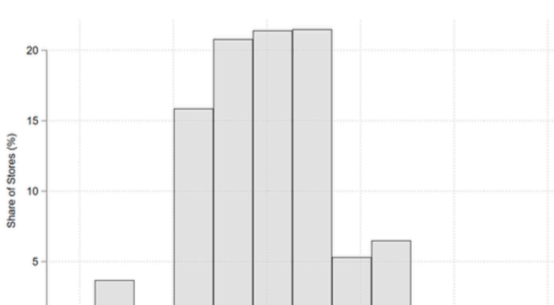


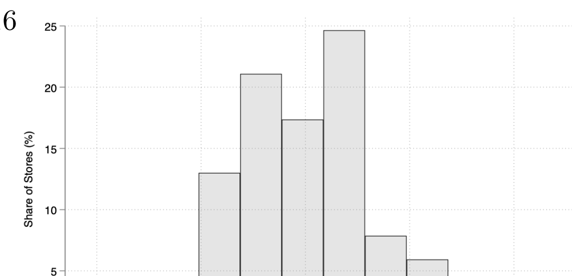
(c) Manager Fixed Effect (Company B)

(a) Paper for Company A



(a) Reproduction for Company A





category	estimate	ci_lower	ci_upper	ci
Percent	0,65	0,49	0,81	[0.49, 0.81]
Value (milli	690469,94	521400,78	859539,06	[521400.78, 859539.06]

(a) Reproduction of Company A

category	estimate	ci_lower	ci_upper	ci
Percent	1,98	1,72	2,23	[1.72, 2.23]
Value (milli	4271929,50	3695996,75	4847862,00	[3.70e+06, 4.85e+06]

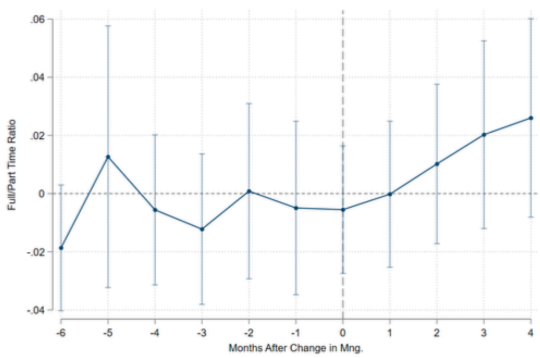
(b) Reproduction of Company B

Table F.1: Sales Revenue Gain per Year with Assortative Matching

	Company A		Company B	
	Estimate	95% CI	Estimate	95% CI
Percent	0.64	[0.48, 0.80]	1.95	[1.67, 2.23]
Value (millions)	0.68	[0.50, 0.85]	4.21	[3.58, 4.84]

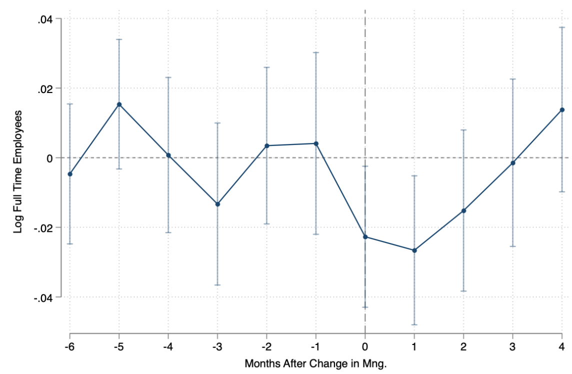
Note: values are in USD for Company A, and GBP for Company B. Confidence intervals are obtained by bootstrapping sample 100 times.

(c) Paper



(c) Full-Time Employment (log)

(a) Manuscript figure G3 (c)



(b) Reproduced figure G3 (c)

10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ `Discrepancy in output` (either figures or numbers in tables or text differ)
- ☐ `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
  - ☐ `Data preparation code missing` should be checked if the code missing seems to be data preparation code
- ☐ `Code not functional` is more severe than a simple bug: it prevented the replicator from completing the reproducibility check

- ☐ `Software not available to replicator` may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ `Insufficient time available to replicator` is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ `Data not available` is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ `Missing README` is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements